

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)

Homework 4 – Ozone Concentration in Hong Kong with Machine Learning

Total Score: 60

Learning Objectives:

- Clean and merge hourly air-quality data with daily meteorological data, and construct a daily ozone target from hourly observations.
- Engineer meteorology-only features for time-series prediction apply time-based train-validation-test splits.
- Implement linear regression and neural-network models and evaluate them using RMSE, MAE, and R^2 .
- Interpret model errors in terms of atmospheric chemistry, emissions, transport, and observation density and communicate the results in a scientific report.

Part 0. Dataset Specifications

You are given a .zip file named hw4_data.zip. Inside the .zip file, there are 10 .csv data files. These files are also available at /lustre/project/eesc4550/shared/hw4.

Data source: air quality data is downloaded from

<https://cd.epic.epd.gov.hk/EPICDI/air/station/?lang=en>, and the meteorological data is downloaded from <https://data.gov.hk/en-datasets/category/climate-and-weather>.

Target: Ozone concentration at Tap Mun, recorded in air_daily.csv

File	Contents	Freq.	Units
air_daily.csv	Hourly air-quality observations (Tap Mun)	Daily	$\mu\text{g}/\text{m}^3$ (CO in $10 \mu\text{g}/\text{m}^3$)

Meteorological Predictors:

File	Variable	Freq.	Units
daily_HKO_CLD_ALL.csv	Daily mean cloud amount	Daily	%
daily_HKO_CLMMAXT_ALL.csv	Daily maximum temperature	Daily	$^{\circ}\text{C}$
daily_HKO_DEW_ALL.csv	Daily mean dew point temperature	Daily	$^{\circ}\text{C}$
daily_HKO_MSLP_ALL.csv	Daily mean sea-level pressure	Daily	hPa
daily_HKO_RF_ALL.csv	Daily total rainfall	Daily	mm
daily_HKO_RH_ALL.csv	Daily mean relative humidity	Daily	%
daily_KP_UV_ALL.csv	Mean UV index (7 a.m.-6 p.m.) at King's Park	Daily	index
daily_TME_PDIR_ALL.csv	Prevailing wind direction at Tap Mun East	Daily	degrees
daily_TME_WSPD_ALL.csv	Mean wind speed at Tap Mun East	Daily	km/h

Air Pollutant Predictors: O_3 , CO and NO_x concentration at Tap Mun in air_daily.csv.

Note: O_3 plays two roles in this assignment. It is the **target** (daily mean O_3 on day t) and it is also a **predictor** (daily mean O_3 on day $t - 1$ as an input to the model) since ozone is **highly persistent** from one day to the next.

For this homework, we consider the period from 2017/09/01 to 2026/08/31.

To code on the Central Cluster, source the gcc.env file, available at

/lustre/project/eesc4550/shared. We have installed the library packages for you.

Part 1. Data Preparation and Feature Engineering

Question 1 (8 pts). Load `air_daily.csv` using `pandas`.

1. Replace all occurrences of "N.A." with `numpy.nan`.
2. Convert the O_3 , NO_x , CO data columns to numeric values using `pd.to_numeric(..., errors='coerce')`. Multiply the data column containing the CO data by 10 for consistency.

Report the following:

- Number of days with valid data, available for all three species.
- Basic statistics: mean, median, standard deviation, minimum, maximum of daily mean O_3 .
- A time-series plot of daily mean O_3 concentrations over the entire period, make sure the time-series line breaks at gaps.

Question 2 (8 pts). Load the daily meteorology `daily_HKO*_ALL.csv` files.

1. For each file, rename the columns:

年/Year, 月/Month, 日/Day, 數值/Value to `year`, `month`, `day`, and the corresponding variable names (e.g. `CLD`, `TMAX`, `DEW`, `MSLP`, `RF`, `RH`, `UV`, `WDIR`, `WSPD`), respectively.

2. Create a proper date column using `pd.to_datetime(df[['year', 'month', 'day']])`.
3. Filter each meteorology file to the same period as the target data.
4. Merge all meteorology files into one daily dataframe using an outer join on date. The result should have one row per day and columns for each meteorological variable.
5. For trace amount of rainfall, treat `RF` of that day as `0` instead of missing.
6. Ensure that days with insufficient data (i.e. any missing meteorological features) are marked as missing.

Report how many days within the time period are available. Compile a summary statistics (mean, median, standard deviation, min, max) for each meteorological variable after cleaning and produce a correlation heatmap of all meteorological variables.

Question 3 (8 pts). Transform wind direction and day-of-year with the following formulae:

$$\text{wind_sin} = \sin\left(\frac{\text{WDIR} \cdot \pi}{180}\right), \text{wind_cos} = \cos\left(\frac{\text{WDIR} \cdot \pi}{180}\right)$$
$$\text{doy_sin} = \sin\left(\frac{2\pi \cdot \text{day_of_the_year}}{365}\right), \text{doy_cos} = \cos\left(\frac{2\pi \cdot \text{day_of_the_year}}{365}\right).$$

This creates four new predictor data columns, namely `WIND_SIN`, `WIND_COS`, `DOY_SIN`, `DOY_COS`.

To build a forecast model, construct a one-day-ahead forecast: the model predicts daily mean O_3 on day t using meteorological and air-quality predictors observed on day $t - 1$. All predictor variables are lagged by one day, so at forecast time only information available up to $t - 1$ is used. After lagging, we drop any day for which the target or any predictor is missing, ensuring that every sample in the final dataset is complete. Next, use a time-based split with proportions 70 % / 15 % / 15 % in chronological order for training dataset, validation dataset, and the test dataset, respectively.

Report the number of valid data samples in the final dataset, the actual split dates. Also, explain why a time-based split is essential for air quality forecasting. Make sure that your data shift is legit as some dropped rows may cause misalignment for the calendar dates.

Part 2. Linear Regression Baseline

Question 4 (10 pts). Fit an Ordinary Least Squares Linear Regression model on the training set using the lagged predictors.

Report on the test set with RMSE, MAE, and R^2 . Produce a **scatter plot** of predicted vs observed daily mean O_3 on the test set, with the 1:1 line for reference.

Part 3. Neural Network Model Baseline

Question 5 (10 pts). Build a feed-forward neural network using `sklearn.neural_network.MLPRegressor` using two hidden layers (64 and 32 neurons), ReLU activation, the Adam optimizer with initial learning rate `1e-3` and early stopping (validation fraction 0.15, max iterations 2000, random state 42).

Next, by adjusting the initial learning rate over `[1e-4, 1e-3, 1e-2, 5e-1]`, select the model with the best validation performance.

Report on the test set with RMSE, MAE, and R^2 . Produce a **scatter plot** of predicted vs observed daily mean O_3 on the test set, with the 1:1 line for reference by using the model with the best validation performance.

Part 4. Model Performance Improvement Challenge

Question 6 (16 pts). Improve your neural-network model so that it achieves at least a **7%** reduction in test-set **MAE** relative to **both** the Q4 OLS baseline and the Q5 MLP baseline.

Rules:

- You may add features (limited to the given meteorological data, NO_x , CO, and their transformations), change model class, ensemble, or stack models.
- You must use the same train/validation/test split for fair comparison.

Report your final model description, test set metrics (RMSE, MAE, R^2), and a scatter plot of the observed vs predicted daily mean ozone concentrations by the final model. Explaining which changes contributed most to the improvement, and which did not help. If your Q6 model does not meet both conditions, report the actual MAE reductions you achieved against Q4 and Q5, and explain with evidence why further improvement was not possible on this dataset.

Note: You will receive partial scores even you cannot reach the 7% target. Any change to the dataset, split, target definition, or methods that cause data leakage will result in **zero** points for this question, regardless of the reported MAE.

Hints: You may try these common approaches: normalization (using the information from the training set only), regularization, multi-day lags of ozone. You may try to allocate some CPUs available on the CUHK Central Cluster to speed up your work.

Submission Guideline: To complete the homework, please submit **TWO** files:

- `hw4_<your_id>.pdf`, which contains the answers to all the **6** questions;
- `hw4_<your_id>.zip`, which contains the Python scripts to produce the included plots and scripts that perform data analysis.

If any AI tool was used in writing or code generation, please include a short note on AI use.

----- End of Homework 4 -----